

Active Planet

EARTH IS A DYNAMIC PLANET that is always changing its form. Heat generated by nuclear reactions deep below the surface creates hugely powerful currents that keep Earth's rocks on the move, triggering earthquakes and volcanic eruptions. Meanwhile, solar energy striking the planet in different ways creates currents in the air, driving the atmospheric turmoil of the weather. This changes with the seasons and from place to place, creating an enormous range of climates and habitats for the most dynamic element of all—life.

DOWN TO THE CORE
Earth formed from iron-rich asteroids that smashed together to build the planet. Early in its history it melted, allowing the heavy iron to sink and create a metallic core. This is surrounded by lighter rock, with the lightest forming Earth's crust. Most of the water on the planet lies in great oceans, and above them is the layer of air that forms the atmosphere.

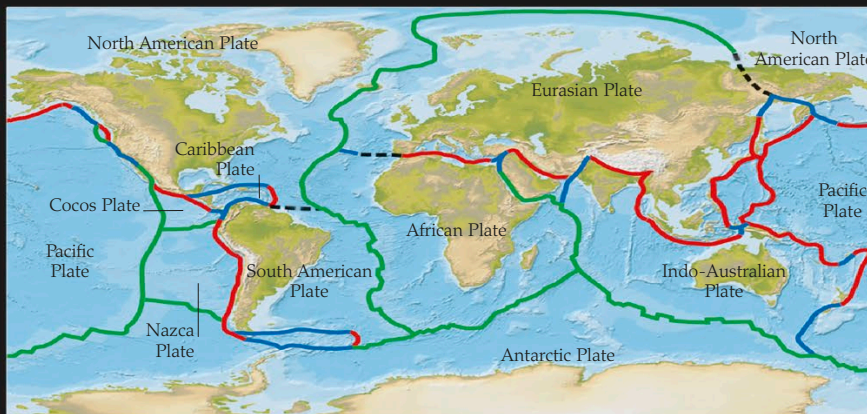
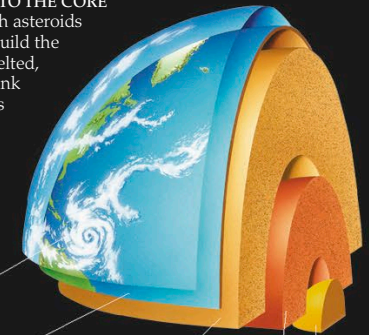
Lower atmosphere,
10 miles (16 km) thick

Crust, 5–45 miles
(8–70 km) thick

Mantle, 1,800 miles
(2,900 km) thick

Liquid outer
core, 1,400 miles
(2,250 km) thick

Solid inner
core, 1,515
miles (2,440
km) across



THE PLATES OF EARTH'S CRUST

Heat generated deep within the planet creates currents in the mobile mantle rock beneath the crust. These currents drag some sections of the cool, brittle crust apart while pushing other parts together, fracturing the crust into separate plates. The biggest of these span oceans and continents, but there are many smaller plates. At their boundaries the plates may be diverging (pulling apart), converging (pushing together), or sliding past each other at transform faults.

Key to map

— Transform fault
--- Uncertain boundary

— Divergent boundary
— Convergent boundary

WHERE MOVING PLATES MEET

The boundaries between the plates are volcanic earthquake zones. The plates move very slowly, pulling apart at divergent boundaries. This allows hot rock below to melt, erupt, and cool to form new crust—especially at the spreading rifts that form mid-ocean ridges. Meanwhile at convergent boundaries, one plate slides beneath another, pushing up mountain ranges and making volcanoes erupt. Other volcanoes erupt over hot spots in the mantle below the crust.

- 1 Continental crust, much thicker than oceanic crust
- 2 Broad basin formed near uplifted area
- 3 Ancient converging boundary, now inactive
- 4 Mountains created when plate boundary was active
- 5 Oceanic crust formed from heavy basalt rock
- 6 Upper mantle, mainly solid but very hot
- 7 Mantle, solid but mobile owing to heat currents
- 8 Spreading rift forming a mid-ocean ridge
- 9 Hot-spot volcano erupting over mantle plume
- 10 Ocean trench marking convergent plate boundary
- 11 Volcano erupting over convergent boundary
- 12 Earthquake zone—one plate grinding under another
- 13 Plates pulling apart, creating a rift valley

